

Evaluation of Asymmetric link-based AODV Routing Protocol for Low Power and Lossy Networks in COOJA Simulator

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Abstract—Route discovery for Point-to-Point (P2P) traffic flows is critical in Low-power and Lossy Networks (LLNs), especially in home and building automation applications. Although the Routing Protocol for Low-Power and Lossy Networks (RPL), standardized by the Internet Engineering Task Force (IETF), is widely used, it struggles to efficiently handle P2P flows, leading to congestion at the Destination Oriented Directed Acyclic Graph (DODAG) root and increased packet delays. The P2P-RPL protocol aims to address this issue, but neither RPL nor P2P-RPL adequately account for asymmetric wireless links in their route computations. AODV-RPL, a protocol draft adopted by the IETF's ROLL working group, offers a reactive P2P route discovery mechanism based on Ad Hoc On-demand Distance Vector Routing (AODV), operating with RPL in its storing mode. This study assesses the performance of AODV-RPL using the Cooja simulator across various network topologies and node densities, considering both symmetric and asymmetric links. Through extensive simulations, it is demonstrated that AODV-RPL outperforms traditional RPL in Packet Delivery Ratio (PDR) and delay performance for numerous source and destination pairs. However, while AODV-RPL generally improves upon traditional RPL for P2P routes, some node pairs remain where the difference in relative hop distance is minimal.

Index Terms—AODV-RPL AODV, DODAG, RPL and P2P links

I. INTRODUCTION

The Internet of Things (IoT) encompasses many constrained devices like sensors and actuators, which interact with their surroundings, connect via networks, and are accessible through the Internet [1–3]. 6LoWPAN [4] is the proposed standard for translating IPv6 packets into IEEE 802.15.4 MAC frames through compression and fragmentation. Developing a robust routing protocol is crucial for improving the performance of memory and power-constrained nodes in networks that

undergo dynamic changes [5]. In response to this need, the Internet Engineering Task Force (IETF) introduced RPL, a proactive IPv6 distance vector routing protocol tailored for Low-power and Lossy Networks [6]. RPL's primary objective is to establish and maintain loop-free routing paths across multi-hop networks, catering to the demands of various real-world IoT applications, often involving multi-point-to-point or point-to-multipoint configurations. However, the assumptions inherent in RPL can be limiting when applied to networks in home and building automation settings, characterized by stringent requirements for message reliability and latency among communicating nodes within the same network. Consider, for instance, a scenario involving remote control operations for home appliances. The inadequacies of RPL in facilitating peer-to-peer (P2P) communication are best elucidated by examining its application in such contexts.

In Low-power and Lossy Networks (LLNs), constrained nodes running the RPL protocol are interconnected by constructing a DODAG. The DODAG root, acting as the LLN border router, initiates the dissemination of DIO control messages to establish a DODAG tailored to a specific instance requirement called Objective Function. Through this process, each node within the LLN can select its parent node, typically opting for the most suitable candidate from a set of neighbouring nodes. Consequently, a path is formed from the nodes within the network through the DODAG root. RPL offers two operational modes to accommodate nodes with varying resource capabilities. In the non-storing mode, utilized when nodes face constraints in processing and memory, all routes are maintained at the DODAG root. Consequently, all peer-to-peer (P2P) communications occur via the DODAG root, leading to

longer routes between peers and radio congestion at the root node. Conversely, in the storing mode, suitable for resourceful nodes, each node in the network retains downward routes to its child nodes. However, these routes are suboptimal, as packets from the originating node traverse upwards until they reach a common ancestor with a downward path to the destination node. A reactive Point-to-Point protocol (P2P-RPL) has been proposed to mitigate such suboptimal routes. While P2P-RPL proves more efficient for P2P traffic in non-storing mode, its address vector usage in storing mode results in significant network overhead.

Furthermore, neither traditional RPL nor P2P-RPL accounts for symmetric links in route computation. AODV-RPL [8] is a reactive protocol proposed for both symmetrical and asymmetrical links to overcome the disadvantages of RPL and P2P-RPL. AODV-RPL creates a temporary paired instance between source and destination nodes to transport data packets. The source node, which needs to send data, initiates an AODV-RPL instance called RREQ-instance by multi-casting DIO with the RREQ message. The destination node, in turn, multi-casts or uni-casts a response message called the RREP message. The RREP message establishes a path from the source to the destination node, and the RREQ message establishes a path from destination to source.

This paper comprehensively evaluates the performance of RPL and AODV-RPL in a dense network of constrained nodes. An extensive set of experiments were conducted, having considered both symmetric and asymmetric links between multiple node pairs in the network.

The paper is organised as follows. Section. II briefly explains the pros and cons of traditional RPL and P2P-RPL routing protocol. Section.III briefly overviews the proposed AODV-RPL protocol with symmetric and asymmetric link design. Section. IV explains implementation. Section. V discuss the experimental results tested in the Cooja simulator with the Contiki operating system. Section. VI completes the paper with the Conclusion and Future work.

II. RELATED WORK

RPL, an IPv6 distance vector routing protocol tailored for Low-Power and Lossy Networks (LLNs), is engineered to accommodate diverse traffic flows within a root-oriented Directed Acyclic Graph (DAG) [6]. In RPL, data packets are required to traverse the root in non-storing mode or a common ancestor in storing mode. However, RPL often encounters significant traffic congestion near the DAG root [7, 9], leading to suboptimal routing paths. To address this issue and discover optimal routes for Peer-to-Peer (P2P) traffic within the RPL framework, a reactive-based P2P-RPL protocol has been introduced [7]. P2P-RPL establishes a temporary DODAG with a designated SourceNode acting as a transient root. Employing an address vector for routing in both storing and non-storing modes, P2P-RPL exhibits slightly reduced efficiency in storing mode. The protocol's implementation and performance evaluation are detailed in [10].

Both RPL and P2P-RPL are designed to operate within a single DODAG framework in networks characterized by symmetric links. However, specific routing demands outlined in the IETF ROLL Working Group for constrained networks [11–14] may necessitate routing metrics and constraints tailored to bidirectional asymmetric links. To address this, [8] introduces the concept of bi-directional asymmetric links for traditional RPL, employing Paired DODAGs where the DAG root (DODAGID) is shared between two Instances. This approach enables the fulfillment of application-specific routing requirements for bidirectional asymmetric links within the basic RPL framework. For constrained networks featuring bi-directional asymmetric links, AODV-RPL introduces P2P route discovery, integrating RPL with a new Mode of Operation (MoP). AODV-RPL employs two multicast messages for asymmetrical links: one from the SourceNode to the TargetNode and another from the TargetNode to the SourceNode. By utilizing AODV-RPL, the necessity for an address vector is obviated for symmetrical links, leading to a significant reduction in control packet size. This study compares RPL with AODV-RPL concerning both symmetric and asymmetric links, analyzing Packet Delivery Ratio (PDR), delay, and number of hops utilizing an LLN border router. Building upon prior research [16] which focused on a 16-node network, experiments are conducted to assess delay and PDR metrics, considering various source and destination pairs.

III. OVERVIEW OF AODV-RPL

Reactive route discovery in AODV-RPL starts with a temporary DAG rooted at the SourceNode. The AODV-RPL Mode of Operation (MoP) creates paired DODAGs, RREQ-Instance and RREP-Instance, during route creation between the SourceNode and TargetNode. Multicasting route control messages from SourceNode to TargetNode creates RREQ-Instance. RREP-Instances are established by multicasting or unicasting route control messages from the TargetNode to the OrigNode. Intermediate routers in the Low-Power and Lossy Network (LLN) join Paired DODAGs based on DIO message rank.

The DIO header of an LLN node sending the AODV-RPL message contains the following:

- S bit: The Symmetric bit, an enhancement to DIO. The AODV-RPL DIO message requires a DIO object Mode of Operation of 5. The AODV-Instance's paired DODAGs' Instance-ID is RPLInstanceID. Device initiating the AODV RREQ-Instance's IPv6 address. AODV-RPL instance node rank.

The DIO base object's S bit is set to '1', indicating a symmetric path from SourceNode to TargetNode. RREQ-Instances arriving over symmetric interfaces with the S bit set to 1 stay set at 1. RREQ-Instance hits intermediate nodes that are uncertain to be symmetric or known to be asymmetric, setting the DIO base object's S bit to 0. Expected Transmission Count (ETX) determines whether the intermediate node link is symmetric (S=1) or asymmetric (S=0). The S bit stays 0 when retransmitted at intermediate nodes (Figure 3). Before broadcasting the RREP-Instance message upstream

to the SourceNode, the TargetNode checks the S bit in the DIO-RREQ-Instance to determine route symmetry. RREQ-Instances transmit application data from the TargetNode to the SourceNode, while RREP-Instances transmit from the SourceNode to the TargetNode.

IV. IMPLEMENTATION

To analyse the behaviour of RPL and AODV-RPL, the specification has been implemented on Contiki [15], a wireless sensor network operating system used and actively developed by a large industrial and academic community. Contiki was chosen for its IPv6 stack and 6LoWPAN capabilities. AODV-RPL implementation was based on its RPL implementation, and the Cooja simulator can simulate it.

Figures 1, 2, and 3 illustrate 16, 38, and 80-node networks from the tests. We suppose Node-1 is an LLN border where LLN traffic is rerouted through a wired backbone. Cooja [15], a wireless sensor network emulator, simulated the networks. Table I lists the experiment setup.

TABLE I
SIMULATION CONFIGURATION IN COOJA

Platform	MSP430 based motes
MAC	CSMA
Network Stack	IPv6/6LoPAN
Radio duty cycle(RDC)	Null RDC
Application	UDP sever/ UDP client
Application data length	40 bytes
Application data rate	Fixed
MAC max retries	3
Mote used	Wismote

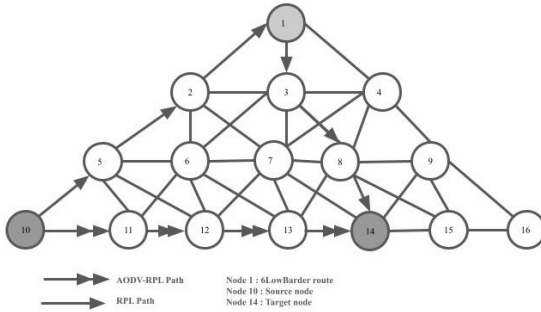


Fig. 1. Topology with 16 nodes.

The Radio medium used in Contiki is Directed Graph Radio Medium (DGRM) because it is easier to configure link states between nodes like RSSI, PRR, LQI, etc. Random SourceNode and TargetNode pairs were considered. Links were made asymmetric between random SourceNode and TargetNode pairs by varying Packet Reception Ratio (PRR) values provided in the DGRM configuration.

With AODV-RPL, a route discovery to the TargetNode was initiated by the SourceNode node, and a route reply initiated by the TargetNode to the SourceNode was recorded. For RPL, a global DAG uses DIO to establish upward routes and the DAO mechanism in the non-storing mode to provide paths

from SourceNode to TargetNode, as described in Section III. The number of hops taken to reach TargetNode, delay experienced and PDR achieved were recorded. Such experiments were repeated for different SourceNode and TargetNode pairs with varying link characteristics. Between nodes like RSSI, PRR, LQI, and so on. Random SourceNode and TargetNode pairs were considered. Links were made asymmetric between random SourceNode and TargetNode pairs by varying Packet Reception Ratio (PRR) values provided in the DGRM configuration.

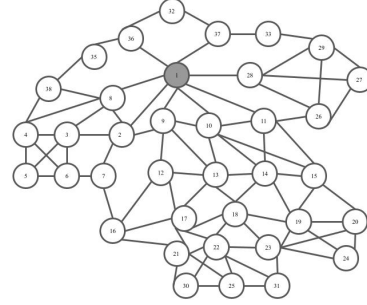


Fig. 2. Topology with 38 nodes.

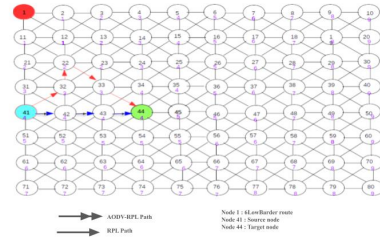


Fig. 3. Topology with 80 nodes.

V. EXPERIMENTAL RESULTS AND OBSERVATION

Experiments were conducted with arbitrarily selected SourceNode and TargetNode pairs. Links were made symmetric and asymmetric by varying PRR values. The data about the number of hops, number of control packets, the path taken by packets, PDR, and the delay incurred for both AODV-RPL for both symmetric and asymmetric links were recorded. The total number of routes is categorised as a measure of the number of hops taken by both protocols between multiple node pairs. Table II shows the percentage of routes that AODV-RPL and RPL use and the number of hops each route takes. The individual performance of both protocols is shown in Table II.

It can be inferred from Table II that, on average, AODV-RPL takes far fewer hops than RPL. About 40% of AODV-RPL routes use four hops, whereas about 38% of routes in RPL use six hops. Another metric considered for comparison is the delay experienced by the packets traversing the network. Another metric considered for comparison is the delay experienced by the packets traversing the network. The histogram

TABLE II
NUMBER OF HOPS USED BY PERCENTAGE OF ROUTES BY RPL AND
AODV-RPL

Number of hops used	Percentage of AODV-RPL routes	Percentage of RPL routes
2	5%	-%
3	25%	7%
4	40%	30%
5	15%	13%
6	15%	38%
7	-	10%

of the delay experienced by these packets is shown in Figure 4. The percentage of packets sent is plotted along the y-axis, and the delay(ms) is along the x-axis. It is observed from Figure 4 that 45% of packets in AODV-RPL experienced a 40 to 50 ms delay, and 29% of the packets experienced a 52- 60 ms delay. In RPL, 37% of packets experienced a delay between 55 to 65 ms, and 25% experienced a delay between 70 to 80 ms. Therefore, It can be deduced from the data that AODV-RPL takes less delay than RPL. One of the reasons for the increased delay in RPL is that the packet must travel to a common ancestor or sometimes to the DODAG root itself. However, AODV-RPL, on the other hand, finds the

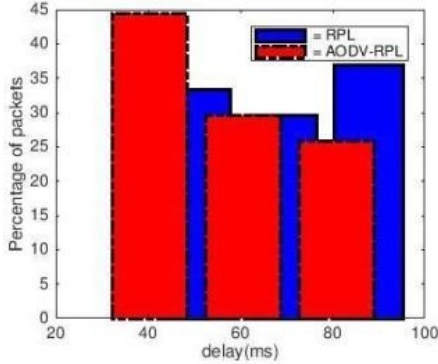


Fig. 4. Summary: delay of RPL and AODV-RPL.

shortest path on demand. As the number of hops increase, the cumulative delay due to re-transmissions between every node-pair also increase. PDR is also observed less in RPL than AODV-RPL, as packets are lost due to congestion at the border router. The experimental results of a single SourceNode and TargetN- ode pair are considered below. The histograms shown in Figure 6 and Figure 5 demonstrate the delay experienced when SourceNode with ID 13 sent packets to TargetNode with ID 10 and subsequent replies from Node-ID 10 to Node-ID 13, respectively. It can be observed that the reply packets take a different path than the sender packet. This is due to the asymmetric nature of the links.

From the above graph, it can be deduced that the delay experienced by RPL is higher than that experienced by AODV-RPL. Another observation is that the path in RPL changes over

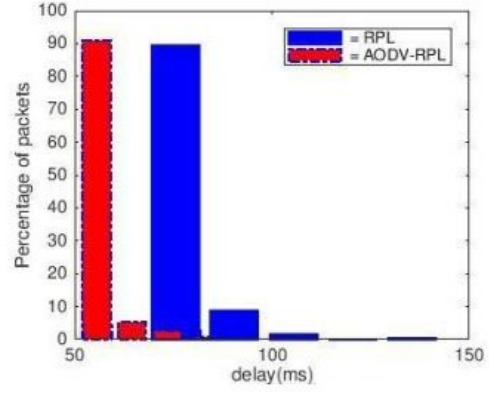


Fig. 5. Packet delay: SourceNode 13 and TargetNode 10 .

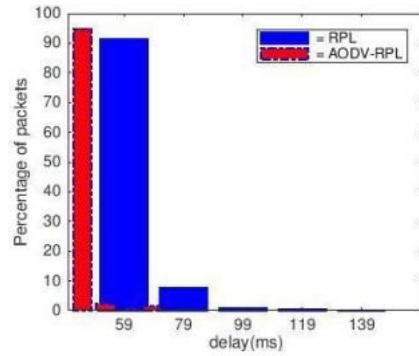


Fig. 6. Packet delay: SourceNode 10 and TargetNode 13 .

time. Initially, DODAG establishes the upward routes; after exchanging a few packets between a node pair, RPL tries to find a better path. This causes a change in the parent and, subsequently, in the path, leading to packet loss. However, AODV-RPL will establish a direct path between the node pairs, and the same path is retained throughout the instance. The change in delay due to symmetric and asymmetric links for the same node-pairs has also been observed.

It noted that the time between originating the route request and receiving the route reply was measured to be 550 milliseconds on average, as the trickle timer is not used for the first RREQ/RREP message. This node received RREQ/RREP, immediately uni-cast/multi-cast RREQ/RREP. RPL with the topology as in Figure 2, and for SourceNode 30 to TargetNode 7, data packets travel through ten nodes following the route as in Figure 7a. The packet from Source node 7 to Target node 30 uses eight nodes, as in Figure 7b. This infers in case of links being asymmetric, requests and packets take different routes, and the number of hops is also different. In the case of AODV-RPL, the path taken for the request packet is 30 → 21 → 16 → 7 and the relay packet takes 7 → 16 → 21 → 30.

The round-trip delay experienced by the packet in the case

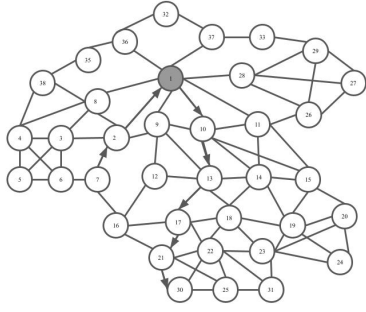


Fig. 7. Packet delay: SourceNode 13 and TargetNode 10 Data packet route between SourceNode 7 to TargetNode 30.

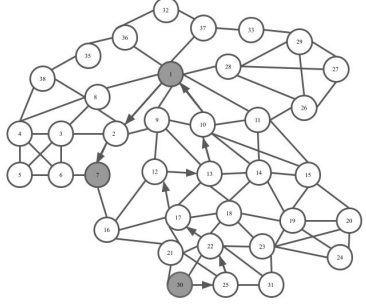


Fig. 8. Data packet route between SourceNode 30 to TargetNode 7.

of symmetric and asymmetric topologies was recorded. Table III shows that the delay experienced in Asymmetric topology for the same source and destination pair is more than that of symmetric because the request and reply packets take different routes. Also, the number of hops for request and reply are extra.

TABLE III
DELAY COMPARISON OF AODV-RPL AND RPL FOR BOTH SYMMETRIC AND SYMMETRIC ROUTES

	RPL Symmetric	RPL Asymmetric	AODV-RPL Symmetric	AODV-RPL symmetric
Max	92	98	58	67
Average	85.5	89.75	52	44.66
standard				
Deviation	9.19	9.77	8.75	11.23

The overall simulation shows that in 7, the percentage of routes passing through RPL-Border-router in the case of AODV-RPL is significantly lower than RPL, which reduces congestion at RPL-Border-router.

TABLE IV
NUMBER OF HOPS USED BY THE PERCENTAGE OF RPL AND AODV RPL

Hop difference	Percentage
less than 2	42%
two or more	58%

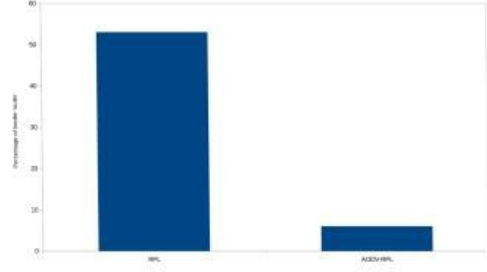


Fig. 9. Percentage of routes using RPL border router.

It can be observed from Figure 4 that 42% of the AODV-RPL and RPL routes experience route length differences of 0 or 1 hops. The investigation provides that if the shortest hop distance between the node pairs is more, then even AODV-RPL can take a route length equal to the RPL route length. For example, in Figure 1, node pairs (10-16) experience six as the shortest hop length, the route length for AODV-RPL and RPL. The route taken by RPL is (10 → 5 → 2 → 1 → 4 → 9 → 16) and the relay packet takes (7 → 16 → 21 → 30) and for AODV-RPL is (10 → 11 → 12 → 13 → 14 → 15 → 16), and in topology 2, between node pair (35-33) AODV-RPL uses (35 → 36 → 32 → 37 → 33). RPL uses (35 → 36 → 1 → 37 → 33), even though RPL uses a border router, when the network is not dense and also when TargetNode is one of its parent towards DODAG root, then route length difference RPL and AODV-RPL are less than two hops. AODV-RPL can be avoided in these situations to reduce AODV-RPL control packet overhead.

From these, a hybrid approach of choosing between RPL and AODV-RPL is recommended depending on application requirements- ment is recommended. A few points to be noted here is that the draft [8] doesn't specify the behaviour of packets before the establishment of routes by AODV-RPL. The first AODV-RPL DIO received is used for path establishment, and the procedure for subsequent DIO is not addressed. The lifetime of the ADOV-RPL established routes is also not specified.

VI. CONCLUSION

The performance of AODV-RPL is evaluated through extensive simulations on the Cooja simulator. A P2P routing protocol draft standard adopted by IETF's roll working group. The message reliability and delay performance comparison between the stored mode of traditional RPL operation and AODV- RPL for P2P flows between various node-pair combinations show that AODV-RPL performs better under different network topologies, node densities, and links exhibiting symmetric and asymmetric properties. While this result is not surprising, It is noted that there are P2P routes where the hop distance difference between routes generated by traditional RPL and AODV-RPL is two or less. This crucial observation leads us to conclude that P2P flows between specific node pairs can reuse the routes already made available by traditional RPL

instead of triggering reactive AODV-RPL routing mechanism with the associated routing protocol overhead if the relative benefits offered by AODV-RPL are not significant.

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REFERENCES

- [1] Lorenzo Vangelista Andrea Zanella, Angelo Castellani and Michele Zorzi. Internet of things reality check. *IEEE Pervasive Computing*, no. 2, pp. 90-91, 16:2, April-June 2017.
- [2] T. Sauter M. Wollschlaeger and J. Jasperneite. The future of industrial communication: Automation networks in the era of the internet of things and industry 4.0. *IEEE Industrial Electronics Magazine*, 11:17–27, March 2017.
- [3] K. J. Singh and D. S. Kapoor. Create your own internet of things: A survey of iot platforms. *IEEE Industrial Electronics Magazine*, 6:57–68, April 2017.
- [4] M. R. Palattella et al. Standardized protocol stack for the internet of (important) things. *IEEE Communications Surveys & Tutorials*, 15:1389–1406, Third Quarter 2013.
- [5] T. H. Lee, H. S. Chiang, L. H. Chang, M. C. Hsieh, H. Wen, and K. M. Yap. Modeling and performance analysis of route-over and mesh under routing schemes in 6lowpan. *2013 IEEE International Conference on Systems, Man, and Cybernetics, Manchester*, 13:3802–3806, 2013.
- [6] T. Winter, P. Thubert, A. Brandt, J. Hui, R. Kelsey, P. Levis, K. Pister, R. Struik, J.P. Vasseur, and R. Alexander. Rpl: Ipv6 routing protocol for low-power and lossy networks. RFC 6550, RFC Editor, March 2012.
- [7] M. Goyal, E. Baccelli, M. Philipp, A. Brandt, and J. Martocci. Reactive discovery of point-to-point routes in low-power and lossy networks. RFC 6997, RFC Editor, August 2013.
- [8] Satish Anamalamudi, Mingui Zhang, Abdur Sangi, Charles Perkins, and S.V.R Anand. Asymmetric aodv-p2p-rpl in low-power and lossy networks (llns). Internet-Draft draft-ietf-roll-aodv-rpl-02, IETF Secretariat, September 2017.
- [9] M. Goyal, E. Baccelli, A. Brandt, and J. Martocci. A mechanism to measure the routing metrics along a point-to-point route in a low-power and lossy network. RFC 6998, RFC Editor, August 2013.
- [10] E. Baccelli, M. Philipp, and M. Goyal. The p2p-rpl routing protocol for ipv6 sensor networks: Testbed experiments. In *SoftCOM 2011, 19th International Conference on Software, Telecommunications and Computer Networks*, pages 1–6, Sept 2011.
- [11] M. Dohler, T. Watteyne, T. Winter, and D. Barthel. Routing requirements for urban low-power and lossy networks. RFC 5548, RFC Editor, May 2009.
- [12] K. Pister, P. Thubert, S. Dwars, and T. Phinney. Industrial routing requirements in low-power and lossy networks. RFC 5673, RFC Editor, October 2009.
- [13] A. Brandt, J. Buron, and G. Porcu. Home automation routing requirements in low-power and lossy networks. RFC 5826, RFC Editor, April 2010.
- [14] J. Martocci, P. De Mil, N. Riou, and W. Vermeylen. Building automation routing requirements in low-power and lossy networks. RFC 5867, RFC Editor, June 2010.
- [15] Adam Dunkels. The contiki operating system.
- [16] K. Hajarathaiah, A. Satish; E. Murali Krishna, S. Abdur Rashid, H.M. Lavanya, “Integration of E-health and Internet of Things,” *Blockchain Technology Solutions for the Security of IoT-Based Healthcare Systems*, Bharat Bhushan@all, 2023, chapter 1.